PUSHFORWARD OF A PSEUDO-COHERENT OBJECT TENSORED WITH A NICE COMPLEX IS PSEUDO-COHERENT WITH BASE-CHANGE COMPATIBLE FORMATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete: base change is quoted from the preceding lemma and m-pseudo-coherence is proved by approximation and truncation

Editorial note. The label lemma-base-change-tensor-pseudo-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- pseudo-coherent objects and m-pseudo-coherence
- approximation of pseudo-coherent complexes by perfect complexes at a given threshold
- cohomological dimension bound N for total direct image over quasi-compact quasi-separated spaces
- the perfect base-change tensor lemma and the base-change compatibility of the total direct image of a tensor product
- truncations and their interaction with derived tensor

Source reference. Original reference 0A1D: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5625–5625. Prerequisite pointer invoked at proof line(s) 6390; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A1H: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6308–6308. Prerequisite pointer invoked at proof line(s) 6392; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08D5: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 790–790. Prerequisite pointer invoked at proof line(s) 6402; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08ES: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3786–3786. Prerequisite pointer invoked at proof

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line(s) 6407; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of finite presentation between algebraic spaces over S. Let $E \in D(\mathcal{O}_X)$ be a pseudo-coherent object. Let $\mathcal{G}^\bullet$ be a bounded above complex of finitely presented $\mathcal{O}_X$-modules, flat over Y, with support proper over Y. Then*

$$K = Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet)$$

is a pseudo-coherent object of $D(\mathcal{O}_Y)$ and its formation commutes with arbitrary base change.

Proof. The statement on base change is Lemma 0A1D. Thus it suffices to show that K is a pseudo-coherent object. This will follow from Lemma 0A1H by approximation by perfect complexes. We encourage the reader to skip the rest of the proof.

The question is étale local on Y, hence we may assume Y is affine. Then X is quasi-compact and quasi-separated. Moreover, there exists an integer N such that total direct image $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$ has cohomological dimension N as explained in Lemma 08D5. Choose an integer b such that $\mathcal{G}^i = 0$ for $i > b$. It suffices to show that K is m-pseudo-coherent for every m. Choose an approximation $P \to E$ by a perfect complex P of $(X, E, m-N-1-b)$. This is possible by Theorem 08ES. Choose a distinguished triangle

$$P \to E \to C \to P[1]$$

in $D_{QCoh}(\mathcal{O}_X)$. The cohomology sheaves of C are zero in degrees $\geq m-N-1-b$. Hence the cohomology sheaves of $C \otimes^{\mathbf{L}} \mathcal{G}^\bullet$ are zero in degrees $\geq m-N-1$. Thus the cohomology sheaves of $Rf_*(C \otimes^{\mathbf{L}} \mathcal{G})$ are zero in degrees $\geq m-1$. Hence

$$Rf_*(P \otimes^{\mathbf{L}} \mathcal{G}) \to Rf_*(E \otimes^{\mathbf{L}} \mathcal{G})$$

is an isomorphism on cohomology sheaves in degrees $\geq m$. Next, suppose that $H^i(P) = 0$ for $i > a$. Then $P \otimes^{\mathbf{L}} \sigma_{\geq m-N-1-a}\mathcal{G}^\bullet \longrightarrow P \otimes^{\mathbf{L}} \mathcal{G}^\bullet$ is an isomorphism on cohomology sheaves in degrees $\geq m-N-1$. Thus again we find that

$$Rf_*(P \otimes^{\mathbf{L}} \sigma_{\geq m-N-1-a}\mathcal{G}^\bullet) \to Rf_*(P \otimes^{\mathbf{L}} \mathcal{G}^\bullet)$$

is an isomorphism on cohomology sheaves in degrees $\geq m$. By Lemma 0A1H the source is a perfect complex. We conclude that K is m-pseudo-coherent as desired. $\square$